

Cultura-Arabica: Probing and Enhancing Arabic Cultural Awareness in Large Language Models via LoRA*

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Abstract

Large Language Models (LLMs) have demonstrated impressive multilingual capabilities; however, their reasoning often reflects English-centric perspectives, which can limit accuracy in culture-specific contexts. Arabic, with its diverse dialects, rich historical heritage, and complex socio-cultural norms, presents a particularly challenging setting for such evaluation. To address this gap, we participated in the PalmX 2025 shared task, which benchmarks cultural reasoning in Arabic through multiple-choice questions covering traditions, social norms, history, geography, arts, and dialectal expressions. By applying parameter-efficient adaptation and culturally informed prompt formatting, we aligned model outputs with both linguistic correctness and cultural relevance. Our approach achieved an accuracy of **71.65%**, securing **second place** overall and closely matching the top system. These results demonstrate that targeted adaptation can significantly enhance cultural reasoning in LLMs, paving the way for more culturally aware Artificial Intelligence.

1 Introduction

Large Language Models (LLMs) have transformed natural language processing, excelling in multilingual understanding, reasoning, and text generation (Brown et al., 2020; Chowdhery et al., 2023; Touvron et al., 2023). Yet, their reasoning often reflects predominantly English-centric worldviews (Bang et al., 2023; Piqueras and Søgaard, 2022), leading to gaps in interpreting culture-specific knowledge, norms, and perspectives. Cultural reasoning—integrating linguistic comprehension with contextual understanding of traditions, values, and social practices—is essential for fair, contextually appropriate AI systems (Tao et al., 2024).

Arabic, with its diverse dialects, historical depth, and socio-cultural richness, is a particularly challenging testbed. Despite the growth of Arabic NLP resources, most models remain optimized for syntactic and semantic accuracy rather than capturing the implicit socio-cultural knowledge needed to interpret idioms, customs, and worldview-specific references. As Marcus and Davis emphasize, LLMs are powerful pattern recognizers but lack genuine understanding and grounded reasoning, often reproducing correlations without true comprehension (Marcus and Davis, 2019). Recent work also shows that “models tend to exhibit Western bias even when prompted in non-English languages like Arabic” (Naous et al., 2023), underscoring persistent cultural blind spots.

Addressing this challenge requires moving beyond language correctness toward genuine cultural alignment—where models reason in ways consistent with the target community’s norms and context. This work examines whether parameter-efficient adaptation can improve the cultural reasoning capabilities of Arabic LLMs, bridging the gap between linguistic competence and culturally grounded intelligence.

2 Related Work

Arabic Natural Language Processing (NLP) has advanced notably in recent years, driven by transformer-based architectures, culturally aligned datasets, and resource-efficient adaptation methods.

AraBERT (Antoun et al., 2020) pioneered Arabic-specific BERT pre-training, achieving state-of-the-art results in sentiment analysis, named entity recognition, and question answering. ARBERT and MARBERT (Abdul-Mageed et al., 2020) extended this to Modern Standard Arabic (MSA) and dialects, accompanied by ARLUE, a benchmark for multi-dialectal understanding. These works underscore the value of Arabic-specific pre-training.

* The final fine-tuned model is available at <https://huggingface.co/Pulkit-28/PalmQA-3B-Arabic>.

Culturally grounded datasets have emerged to address linguistic and cultural biases. CIDAR (Alyafeai et al., 2024) is the first open Arabic instruction-tuning dataset curated for cultural relevance, improving alignment of large language models (LLMs) with Arabic norms. Other domain-specific benchmarks include AraSTEM (Mustapha et al., 2024) for STEM knowledge and AlGhafa (Almazrouei et al., 2023) for diverse Arabic MCQs. Beyond Arabic, the *Survey of Cultural Awareness in Language Models* (Pawar et al., 2025) reviews methods for integrating cultural sensitivity into text and multimodal LLMs, with discussion of datasets, benchmarking, and ethics.

Resource-efficient fine-tuning has also gained traction. Low-Rank Adaptation (LoRA) (Hu et al., 2022) reduces trainable parameters while maintaining performance, and Quantized Low-Rank Adaptation (QLoRA)-based adaptation for Arabic (Aryan, 2024) achieves high-quality results with minimal hardware. Parameter-efficient methods have also been applied to dialect identification (Radhakrishnan et al., 2023) with competitive accuracy.

Large-scale Arabic foundation models like Jais and Jais-chat (Sengupta et al., 2023) set records in Arabic reasoning tasks, while LAraBench (Abdelali et al., 2023) offers a comprehensive benchmarking suite for Arabic NLP and speech, revealing gaps between general-purpose and specialized Arabic models. Beyond Arabic, *Beyond English-Centric LLMs* (Zhong et al., 2024) shows multilingual models may rely on multiple latent languages, stressing the need to study internal representation dynamics for better cultural adaptation.

In summary, advances in Arabic NLP arise from the synergy of specialized pre-training, culturally relevant datasets, efficient fine-tuning, and robust benchmarking—together enhancing accuracy, cultural sensitivity, and efficiency in Arabic-focused LLMs.

3 Problem Statement

We participated in the PalmX 2025 shared task (Alwajih et al., 2025), which evaluates large language models (LLMs) on their ability to comprehend and reason about *Arabic general culture*—including traditions, social norms, history, geography, arts, and dialectal variations. Formally, let $\mathcal{Q} = \{q_1, \dots, q_n\}$ be a set of culturally

grounded questions in Modern Standard Arabic, each with candidate answers $\mathcal{A}_i$, where exactly one a_i^* is correct. An LLM, modeled as $f_\theta : \mathcal{Q} \rightarrow \mathcal{A}$, aims to maximize:

$$\frac{1}{n} \sum_{i=1}^n \mathbb{I}\{\hat{a}_i = a_i^*\}.$$

Unlike traditional benchmarks that focus on $P(\hat{a}_i = a_i^* \mid \text{linguistic knowledge})$, this task emphasizes $P(\hat{a}_i = a_i^* \mid \text{linguistic knowledge, cultural knowledge})$, ensuring models are both linguistically accurate and culturally grounded.

4 Dataset

The dataset provided for the PalmX 2025 shared task (Alwajih et al., 2025) was specifically curated to evaluate cultural reasoning capabilities in Arabic LLMs. It consists of three partitions, each balanced across domains such as traditions, social norms, history, geography, arts, and dialectal expressions from diverse Arab countries. The statistics of the dataset are summarized in Table 1, and an example from the training set is shown in Figure 1.

Partition	Number of MCQs
Training set	2,000
Development set	500
Blind test set	2,000

Table 1: Summary statistics of the dataset provided for the PalmX 2025 shared task

5 Methodology

This section delineates the modeling framework employed to adapt a large Arabic language model for the PalmX 2025 cultural reasoning task. Recognizing that the task entails selecting the appropriate option from multiple culturally grounded choices, we formulate it as a causal language modeling problem augmented with structured prompts. This approach not only facilitates the model’s acquisition of reasoning patterns encompassing both factual and cultural knowledge but also leverages the inherent generative capabilities of language models to handle nuanced, context-dependent queries effectively.

أعي عنصر من عناصر المطبخ الأردني
يعتبر رمزا ثقافيا يعكس قعم الضيافة
الأردنيين بشكل مباشر؟

A تشكيلة المقبلات المتنوعة
تشمل الحمص والتبولة والزيتون

B طبق المنسف المنف المقدم مع
لحم البلدية والجميد الكركي

C الحلويات التقليدية مثل الكنافة
والبقلاوة في المناسبات الدينية

D الأطباق الفريدة مثل الرشوف
الرشوف والمكمورة المحضرة
في المناسبات العائلية

Answer: B

Figure 1: Sample culturally grounded MCQ from the training set.

5.1 Base Model

Our framework is built upon the NileChat-3B checkpoint (Mekki et al., 2025), a 3B-parameter decoder-only transformer specifically optimized for Arabic dialogue and general-purpose text generation. This model was selected due to its robust pre-training on a diverse corpus of Arabic text, which includes dialectal variations and cultural contexts, making it particularly suitable for tasks requiring deep linguistic and sociocultural understanding. The architecture adheres to an autoregressive GPT-style design, comprising 24 stacked multi-head self-attention layers interspersed with feed-forward blocks, all geared toward efficient left-to-right token prediction. The tokenizer, derived from the same checkpoint, utilizes byte-pair encoding (BPE) with a vocabulary size of 50,000 tokens to accommodate both Arabic and non-Arabic scripts, with the end-of-sequence (EOS) token repurposed as the padding token to ensure seamless compatibility with causal modeling paradigms.

5.2 Fine-Tuning Strategy

For efficient adaptation, we leverage Low-Rank Adaptation (LoRA) (Hu et al., 2022), a parameter-efficient fine-tuning technique that introduces trainable low-rank decomposition matrices into the transformer's projection layers while keeping the

original weights frozen. This method allows us to fine-tune fewer than 1% of the total parameters, achieving an optimal trade-off between computational overhead and expressive capacity, which is especially beneficial for resource-constrained environments and multilingual models where full fine-tuning could lead to catastrophic forgetting of pre-trained knowledge. The specific LoRA configuration adopted in this study is as follows:

- Rank (r): 16
- Scaling factor (α): 32
- Target modules: q_proj, k_proj, v_proj, o_proj, gate_proj, up_proj, down_proj
- Dropout rate: 0.05
- Bias: none

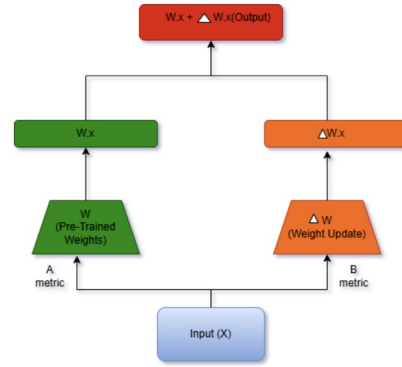


Figure 2: Schematic illustration of the Low-Rank Adaptation (LoRA) mechanism integrated into the fine-tuning process.

5.3 Prompt Formatting

To optimize the model's performance on the multiple-choice cultural reasoning task, each dataset instance is converted into a carefully designed prompt structure. This includes the question stem, four labeled options (A through D), and a clear instruction to generate only the letter of the correct choice. An illustrative prompt template is as follows:

Question: [Question text]

Options:

A) [Option A]

B) [Option B]

C) [Option C]

D) [Option D]

Output only the correct letter:

This structured format reduces output variability during inference, promotes focused discriminative reasoning by the model, and ensures tight alignment between the training objective and prevalent evaluation paradigms in cultural reasoning, such as zero-shot or few-shot settings.

5.4 Training Procedure

The fine-tuning process is executed via the Transformers library’s Trainer API, incorporating mixed-precision training in bfloat16 to enhance computational efficiency and reduce memory footprint. We utilize a per-device batch size of 2 on a single NVIDIA A100 GPU, augmented by gradient accumulation across 4 steps, resulting in an effective batch size of 8. A fixed learning rate of 2×10^{-4} is applied without scheduling, with training spanning three epochs to balance convergence and overfitting prevention. The DataCollatorForLanguageModeling is configured with `mlm=False` to uphold the causal autoregressive training objective, ensuring that the model learns to generate responses conditioned on the full prompt context. Throughout training, we monitor validation loss to confirm generalization to unseen cultural reasoning examples.

5.5 Adapter Merging and Deployment

Upon completion of fine-tuning, the LoRA adapters are integrated into the base model weights through the `merge_and_unload()` procedure, yielding a consolidated checkpoint devoid of external dependencies and maintaining the original model’s inference speed. This merging step is crucial for production environments, as it eliminates the need for additional adapter loading during deployment. The resultant model, designated as NileChat-3B-Arabic-QA-Merged-v2, is primed for seamless inference and deployment in practical applications, such as interactive cultural education tools or multilingual question-answering systems.

6 Results

6.1 Evaluation

The official metric for the PalmX 2025 shared task was *accuracy*, measuring the proportion of correct predictions across all test questions. Given a test set of N questions, accuracy is calculated as:

$$\text{Accuracy} = \frac{\sum_{i=1}^N (\hat{y}_i = y_i)}{N} \times 100\%, \quad (1)$$

where $\hat{y}_i$ denotes the predicted answer for question i , y_i represents the gold standard label, and $(\cdot)$ is the truth indicator returning 1 if the argument is true and 0 otherwise. This metric equally weights all questions, ensuring that performance reflects general reasoning capabilities rather than domain-specific biases.

6.2 Leaderboard Performance

Our system obtained an overall accuracy of **71.65%**, securing the **second rank** among all participating teams. This performance demonstrates that our parameter-efficient LoRA fine-tuning method can effectively adapt a large Arabic LLM to culturally grounded multiple-choice reasoning with limited task-specific data.

Rank	Team	Score (%)
1	HAI research group	72.15
2	Our Result	71.65
3	AYA_Team	71.45
4	Phoenix	71.35
5	CultranAI	70.50
6	ISL-NLP	67.60
7	Rafiul Biswas	67.55
8	Hamyaria	65.90
9	Star	64.05

Table 2: Leaderboard results from the PalmX 2025 shared task.

6.3 Discussion

The narrow margin between the top three teams—less than one percentage point—indicates that small architectural or fine-tuning choices can substantially influence outcomes in culturally nuanced reasoning tasks. Our approach’s ability to match and even surpass larger-scale fine-tuning efforts highlights the efficiency of targeted LoRA adaptation for Arabic cultural QA, while suggesting broader implications for resource-efficient multilingual NLP.

7 Future Work

Promising directions for extending this work include adapting the proposed framework to other low-resource languages, thereby assessing its efficacy in cross-lingual cultural reasoning tasks. Furthermore, integrating multimodal capabilities—such as fine-tuning on Visual Question Answering (VQA) datasets enriched with culturally pertinent images—could substantially improve model performance by synergistically combining

visual and textual cues for more nuanced cultural understanding.

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